

ADITYA SHIRSATRAO

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Solapur, Maharashtra, India

EDUCATION

N. K. Orchid College of Engineering & Technology

Solapur, Maharashtra

Bachelor of Technology in Artificial Intelligence and Data Science – CGPA: 8.5/10

Aug. 2023 – Present (Expected 2027)

Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, Machine Learning, Neural Networks, OOP.

EXPERIENCE

Fullstack Web Development Intern

Unified Mentor Pvt. Ltd

Backend, Distributed Systems & AI-Integrated Products

Jul. 2025 – Jan. 2026

- Architected an **AI-powered code review platform** processing GitHub pull requests asynchronously; integrated Kafka message queues and Redis caching, reducing analysis latency by **45%** through parallel distributed workers and AST-based code search.
- Engineered a **real-time collaborative IDE** with sub-100ms sync using WebSockets, Socket.IO, Redis pub/sub, and CRDT-based operational transforms.
- Built a **distributed ML inference platform** with autoscaling endpoints, model versioning, and Prometheus monitoring; improved throughput by **60%** under concurrent load via request batching and inference caching.

AI/ML Lead

Google Developer Groups of Campus (GDG-NKOCET)

Community Leadership & Mentorship

Sep. 2025 – Present

- Lead AI/ML initiatives and technical workshops for a **200+ student** community; mentored juniors on system design, Git/GitHub workflows, and production ML engineering.

PROJECTS

Order Matching Engine | C++17, Price-Time Priority, GTC/IOC/FOK, FastAPI, WebSockets | Code

Jun. 2026 – Jul. 2026

- Engineered a **dependency-free C++17 limit order book** with price-time priority and FIFO per level, supporting LIMIT/MARKET/STOP/STOP-LIMIT and GTC/IOC/FOK semantics with deterministic integer quantities.
- Benchmarked matching throughput at **10.9M orders/sec** (0.09µs/order); exposed a trade tape + depth feed over a FastAPI/WebSocket bridge; CI builds & tests on every PR.

SentinelX – AI Threat Detection Platform | FastAPI, Redis, DistilBERT, Celery, Redis, PostgreSQL | Code

May 2026

- Built a production-grade, multi-channel threat-detection backend (email/SMS/voice) with a weighted 4-signal risk engine and async Celery + Redis task queues.
- Integrated HuggingFace DistilBERT + OpenAI Whisper STT; shipped JWT auth, RBAC, and Docker Compose; **1st Place** Hack to Future 3.0 (600+ teams, INR 50,000).

Tracelify – Error Tracking & Observability SDK | Python, FastAPI, React, PostgreSQL, CI/CD | Code

Apr. 2026

- Designed and published a multi-language error-tracking SDK (PyPI `tracelify-sdk`) with native Python, Java, Kotlin, Go, C++, JavaScript clients.
- Built a multi-tenant FastAPI backend with Redis caching and LLM-powered error reports; **1st Runner-Up** Orchathon 2K26 (800+ teams).

OPEN SOURCE

- Merged **50+ pull requests** into high-profile open source repos including **Google (OSS-Fuzz)**, **Microsoft (VibeVoice)**, **Apache (DataFusion)**, and **Hedera Hashgraph (Hiero SDK)**.
- Hiero SDK (Hedera):** Refactored the EVM transaction builder to support direct alias conversion, cutting boilerplate by **30%** and eliminating type redundancy errors (PR #1030).

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript (ES6+), TypeScript, SQL

Frameworks: FastAPI, Node.js, Express.js, React.js, Next.js, Flask

Infrastructure: Docker, Kubernetes, Redis, Kafka, GitHub Actions, AWS (EC2/S3/Lambda), Linux

Core Concepts: DSA (366+ LeetCode), Distributed Systems, Microservices, CI/CD, Asynchronous Programming, RESTful APIs

ACHIEVEMENTS & CERTIFICATIONS

Competitive Programmer: 366+ LeetCode problems solved (104 Easy, 176 Medium, 86 Hard) + 50 SQL problems.

IEEE First-Author Publication presented at ICCTWC 2026; **1st Place** at Hack to Future 3.0 (600+ teams) and **1st Runner-Up** at Orchathon 2K26 (800+ teams).

Certifications: JPMorgan Chase & Electronic Arts Software Engineering Virtual Experience; NPTEL Programming in Java (Elite+Gold, 91/100).